

Security Experiment Worksheet

Grand Marina Hotel — TLS Security Assessment

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Experiment 1: Eavesdropper Test

Question: Can someone spy on our sensor data?

Scenario	What You Observed
Without TLS (Part A)	Saw 5 messages with device IDs and pressure readings and time stamps all in plain text
With TLS (Part B)	TLS error- no messages came through

Conclusion (Did TLS block eavesdropping?):

Yes, TLS blocked eavespropping.

Screenshot:

```
[kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % mosquito_sub -h localhost -p 1883 -t "grandmarina/#" -v
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 1, "pressure_psi": 82.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:00"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 2, "pressure_psi": 83.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:01"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 3, "pressure_psi": 84.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:02"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 4, "pressure_psi": 85.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:03"}
grandmarina/sensors/test/telemetry {"device_id": "test-sensor-001", "message_num": 5, "pressure_psi": 86.5, "flow_rate_gpm": 45.0, "timestamp": "14
:39:04"}
^C
ub -h localhost -p 8883 -t "grandmarina/#" -v
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % mosquito_s

Error: Protocol error
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % █
```

Experiment 2: Certificate Test

Question: What happens when certificates are wrong?

Scenario	Result	Error Message (if any)
A: Correct certificates	SUCCESS / FAILED	
B: Wrong CA	SUCCESS / FAILED	FAILED: [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: self-signed certificate in certificate chain (_ssl.c:1081)
C: No verification	SUCCESS / FAILED	

Conclusion (What does this tell you about certificate validation?):

This shows that certificate validation actually matters. When everything is set up correctly, it connects fine. But if the CA is wrong, it fails because it cannot verify the server. By turning off the verification it still connects, however it's not secure since it's not checking who's talking to it.

Screenshot:

```
Done publishing!
[kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode connect --tls on

=====
Connection Test
TLS: ON
CA Certificate: certs/ca.pem
=====

SUCCESS: Connected to broker!
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode generate-wrong-ca

Generating wrong CA certificate for testing...
Saved: certs/wrong-ca.pem
To test: Use this as ca_certs in your client
[kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode test-wrong-ca
Testing connection with wrong CA...

=====
Connection Test
TLS: ON
CA Certificate: certs/wrong-ca.pem
=====

FAILED: [SSL: CERTIFICATE_VERIFY_FAILED] certificate verify failed: self-signed certificate in certificate chain (_ssl.c:1081)
[kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode connect --tls on --no-ca

=====
Connection Test
TLS: ON
CA Certificate: NONE
=====

SUCCESS: Connected to broker!
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % █
```

Experiment 3: Speed Test

Question: Does encryption slow things down?

Metric	TLS OFF	TLS ON
Average latency	52.75 ms	52.55 ms
Min latency	0.65 ms	0.56 ms
Max latency	105.80 ms	106.17 ms

Calculated TLS overhead: -.38%

Formula: TLS Overhead = (TLS_avg - NoTLS_avg) / NoTLS_avg x 100%

Conclusion (Is this overhead acceptable for Grand Marina's sensors?):

Yes, there is no big slow down when using TLS.

Screenshot:

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode latency --tls off --count 50
```

```
=====
Latency Test
TLS: OFF
Messages: 50
=====

Sent 10/50 messages...
Sent 20/50 messages...
Sent 30/50 messages...
Sent 40/50 messages...
Sent 50/50 messages...

=====
Latency Results (TLS OFF)
=====
Messages sent: 50
Average latency: 52.75 ms
Min latency: 0.65 ms
Max latency: 105.80 ms
Std deviation: 51.72 ms
=====
```

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % python3 experiment_runner.py --mode latency --tls on --count 50
```

```
=====
Latency Test
TLS: ON
Messages: 50
=====

Sent 10/50 messages...
Sent 20/50 messages...
Sent 30/50 messages...
Sent 40/50 messages...
Sent 50/50 messages...

=====
Latency Results (TLS ON)
=====
Messages sent: 50
Average latency: 52.55 ms
Min latency: 0.56 ms
Max latency: 106.17 ms
Std deviation: 51.55 ms
=====
```

```
kaylacolletti@Kaylas-MacBook-Pro hydroficient-project % █
```

Experiment 4: Stress Test

Question: Can the system handle Grand Marina's load?

Rate	TLS OFF	TLS ON
10 msg/sec (Normal)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
25 msg/sec (Moderate)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
50 msg/sec (Emergency)	SUCCESS / DEGRADED	SUCCESS / DEGRADED
100 msg/sec (Beyond req.)	SUCCESS / DEGRADED	SUCCESS / DEGRADED

Conclusion (Can the TLS-secured system handle Grand Marina's load requirements?):

Yes, the TLS-secure system can still handle the Grand Marina's load. Turning TLS on really doesn't slow anything down, and it stays stable across normal, moderate, and even higher traffic levels.

Screenshot:

TLS OFF:

```
=====
Stress Test Results (TLS OFF)
=====
Target rate: 10 msg/sec
Actual rate: 9.7 msg/sec
Messages sent: 300
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)
=====
```

```
=====
Stress Test Results (TLS OFF)
=====
Target rate: 25 msg/sec
Actual rate: 24.2 msg/sec
Messages sent: 750
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)
=====
```



```
=====
Stress Test Results (TLS OFF)
=====
Target rate: 50 msg/sec
Actual rate: 48.4 msg/sec
Messages sent: 1500
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)
=====
```

```
=====
Stress Test Results (TLS OFF)
=====
Target rate: 100 msg/sec
Actual rate: 96.8 msg/sec
Messages sent: 3000
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)
=====
```

TLS ON:

```
=====
Stress Test Results (TLS ON)
=====
Target rate: 10 msg/sec
Actual rate: 9.7 msg/sec
Messages sent: 300
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)
=====
```

=====

Stress Test Results (TLS ON)

=====

Target rate: 25 msg/sec
Actual rate: 24.2 msg/sec
Messages sent: 750
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)

=====

=====

Stress Test Results (TLS ON)

=====

Target rate: 50 msg/sec
Actual rate: 48.4 msg/sec
Messages sent: 1500
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)

=====

=====

Stress Test Results (TLS ON)

=====

Target rate: 100 msg/sec
Actual rate: 96.8 msg/sec
Messages sent: 3000
Errors: 0
Success rate: 100.0%
Status: SUCCESS (achieved target rate)

=====